Changepoint detection in network measurements

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Fundamental problem:

- Predict next value in time series.

Applications:

- Protocol parameters (timeouts).
- Resource selection, scheduling.
- User feedback.

Two kinds of prediction:

- Single value prediction.
- Predictive distribution.
  - Summary stats.
  - Intervals.
  - $P(error > thresh)$
  - $E[cost(error)]$

If we assume stationarity, life is good.

- Accumulate data indefinitely.
- Predictive distribution = observed distribution.

Non-stationary models:

- Trends + noise.
- Level + changepoint + noise.

Network performance:

- Some trends (accumulating queue).
- Many abrupt changepoints.
  - Beginning and end of transfers.
  - Routing changes.
  - Hardware failure, replacement.

Prediction with *known* changepoints:

- Use data back to the latest changepoint.
- Less accurate immediately after.

Prediction with *probablistic* changepoints.

$$P(i) = \text{prob of a changepoint after point } i$$

Example:

- 150 data points.
- $P(50) = 0.7$
- $P(100) = 0.5$

How do we generate a predictive distribution?

Two steps:

- Derive $P(i+)$ = prob that i is the last changepoint
- Compute weighted mix going back to each i.

Example:

- $P(50) = 0.7 \qquad P(100) = 0.5$
- $P(50+) = 0.35 \quad P(100+) = 0.5$
- Plus 0.15 chance of no changepoint.

Predictive distribution =

$$
\begin{aligned}
0.50 \quad &\cdot \quad edf\,(100, 150) \oplus \\
0.35 \quad &\cdot \quad edf\,(50, 150) \oplus \\
0.15 \quad &\cdot \quad edf\,(0, 150)
\end{aligned}
$$

So how do we generate the probabilities $P(i+)$?

Three steps:

- Bayes' theorem.
- Simple case: we know there is 1 changepoint.
- General case: unknown # of changepoints.

Bayes' theorem (diachronic interpretation)

$$P(H|E) = \frac{P(E|H)}{P(E)} P(H)$$

- H is a hypothesis, E is a body of evidence.
- $P(H|E)$: posterior
- $P(H)$: prior
- $P(E|H)$ is usually easy to compute.
- $P(E)$ is often not.

Unless we have a suite of exclusive hypotheses.

$$P(E) = \sum_{H_i \in S} P(E|H_i)P(H_i)$$

In that case life is good.

- If we know there there is exactly one changepoint in an interval...
- ...then the $P(i)$ are exclusive hypotheses,
- and all we need is $P(E|i)$.

Which is pretty much a solved problem.

What if the # of changepoints is unknown?

- $P(i)$ are no longer exclusive.
- But the $P(i+)$ are.
- And we can write a system of equations for $P(i+)$.

$$P(i^+) = P(i^+|\oslash)\ P(\oslash) + \sum_{j<i} P(i^+|j^{++})\ P(j^{++})$$

- $P(j^{++})$ is the prob that the second-to last changepoint is at i.
- $P(i^+|j^{++})$ reduces to the simple problem.
- $P(\oslash)$ is the prob that we have not seen two changepoints.
- $P(i^+|\oslash)$ reduces to the simple problem (plus).

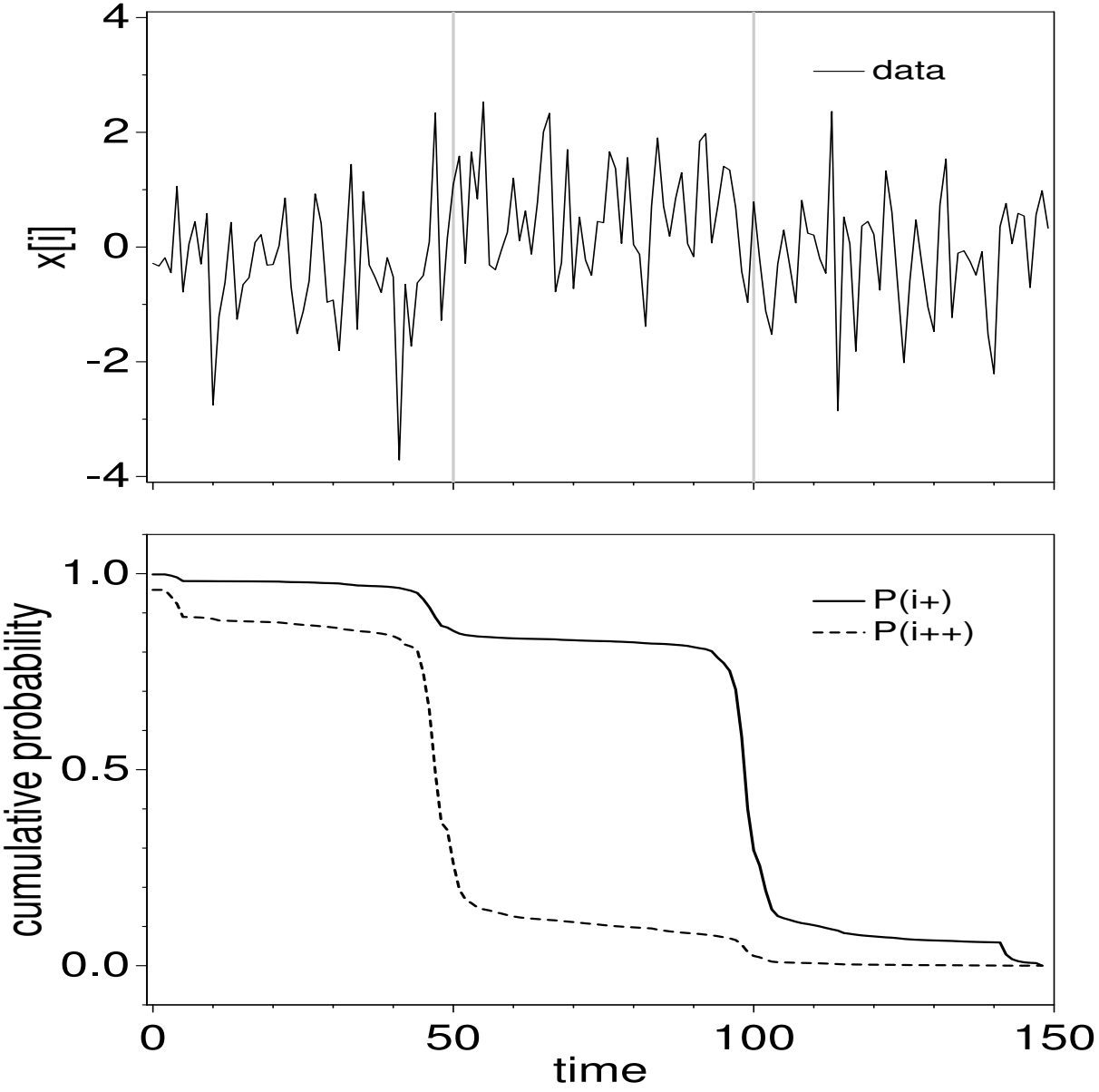
Great, so what's $P(j^{++})$?

$$P(i^{++}) = \sum_{k>i} P(i^{++}|k^{+})\ P(k^{+})$$

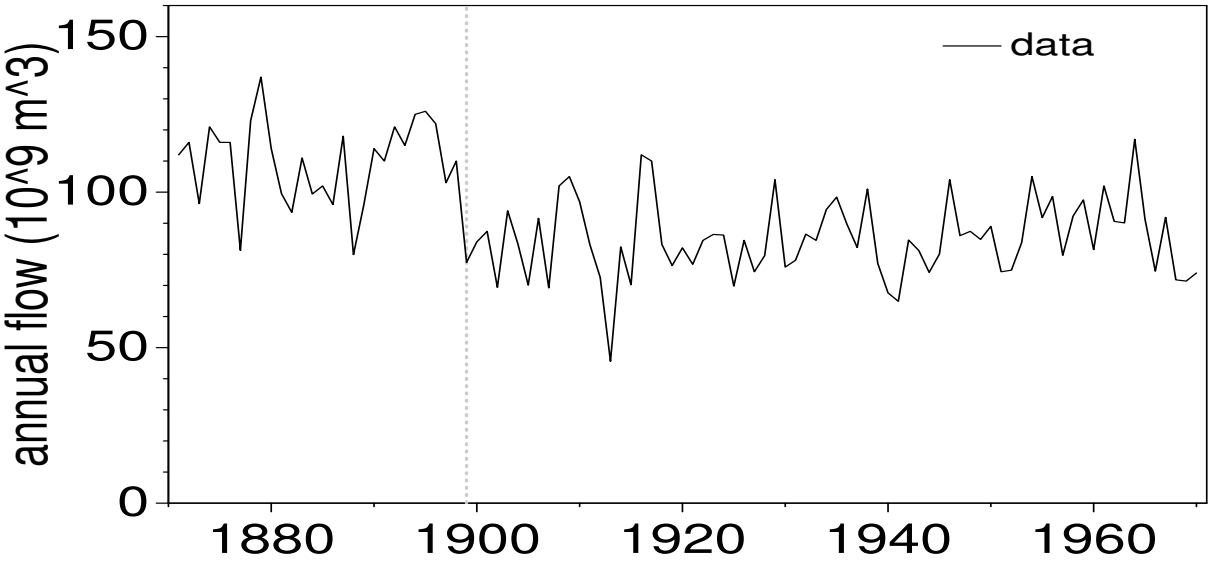
- $P(i^{++}|k^{+})$ is just $P(i^{+})$ computed at time k.
- So we can solve for $P(i^{+})$ in terms of $P(i^{++})$.
- And $P(i^{++})$ in terms of $P(i^{+})$.
- Calling Dr. Jacobi!

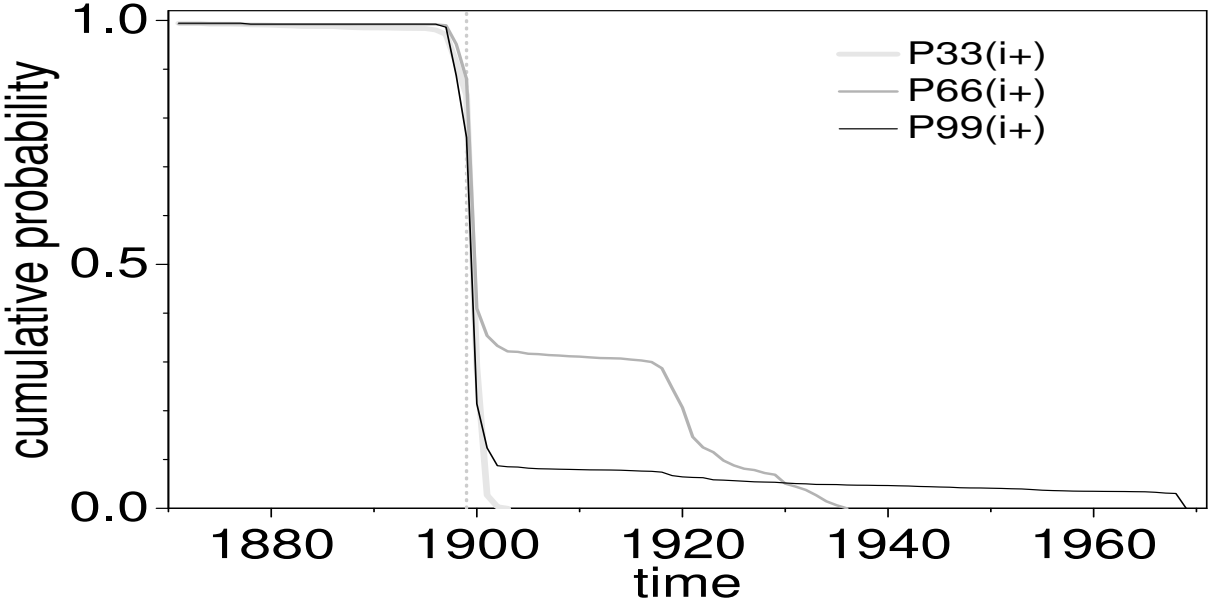
Implementation:

- Need to keep $n^2/2$ previous values.
- And $n^2/2$ summary statistics.
- And it takes n^2 work to do an update.
- But, we only have to go back two changepoints,
- ...so we can keep n small.

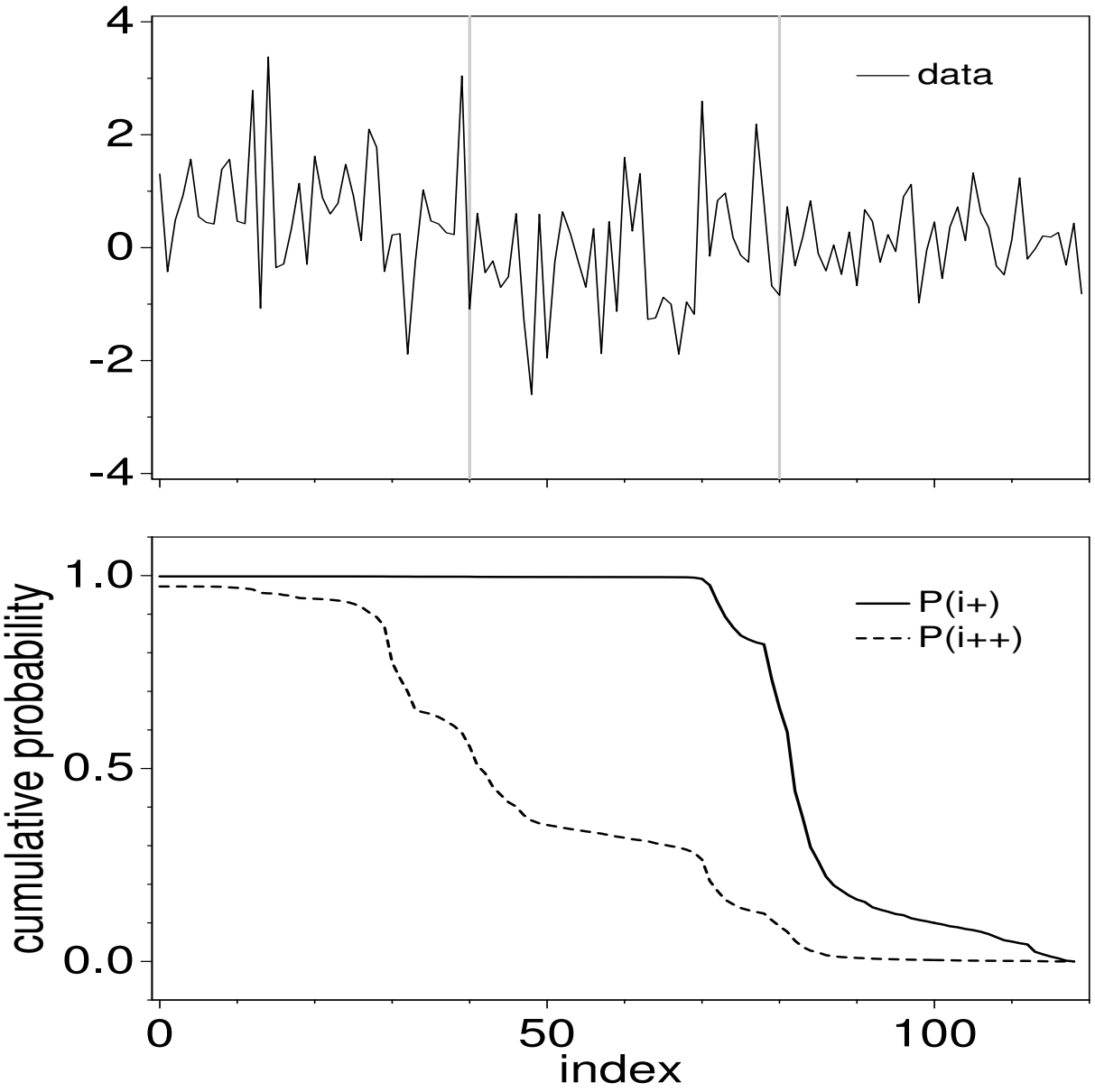


- Synthetic series with two changepoints.
- $\mu = -0.5, 0.5, 0.0$
- $\sigma = 1.0$
- $P(\oslash) = 0.04$





- The ubiquitous Nile dataset.
- Change in 1898.
- Estimated probs can be mercurial.



- Can also detect change in variance.
- $\mu = 1, 0, 0$
- $\sigma = 1, 1, 0.5$
- Estimated $P(i^{+})$ is good.
- Estimated $P(i^{++})$ less certain.

- Qualitative behavior seems good.
- Quantitative tests:
  - Compare to GLR for online alarm problem.
  - Test predictive distribution with synthetic data,
  - ... and with real data.

Online alarm problem:

- Observe process in real time.
- μ_0 and σ known.
- τ and μ_1 unknown.
- Raise alarm when $f(data) > thresh$.
- Minimize delay.
- Minimize false alarm rate.

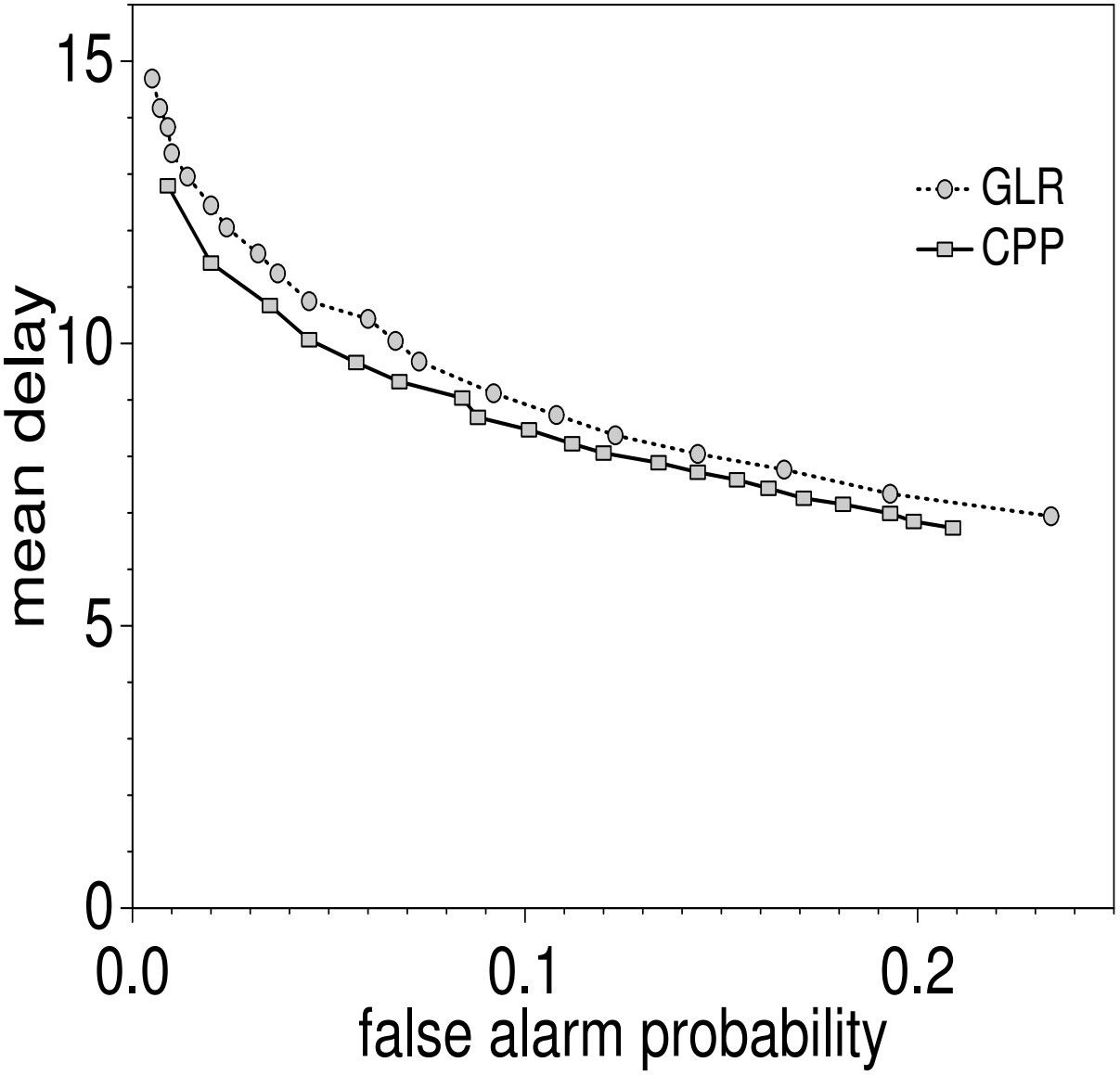
GLR = generalized likelihood ratio.

- Compute decision function g_k.
- $E[g_k] = 0$ before the changepoint,
- ... increases after.
- Alarm when $g_k > h$.
- GLR is optimal when μ_1 is known.

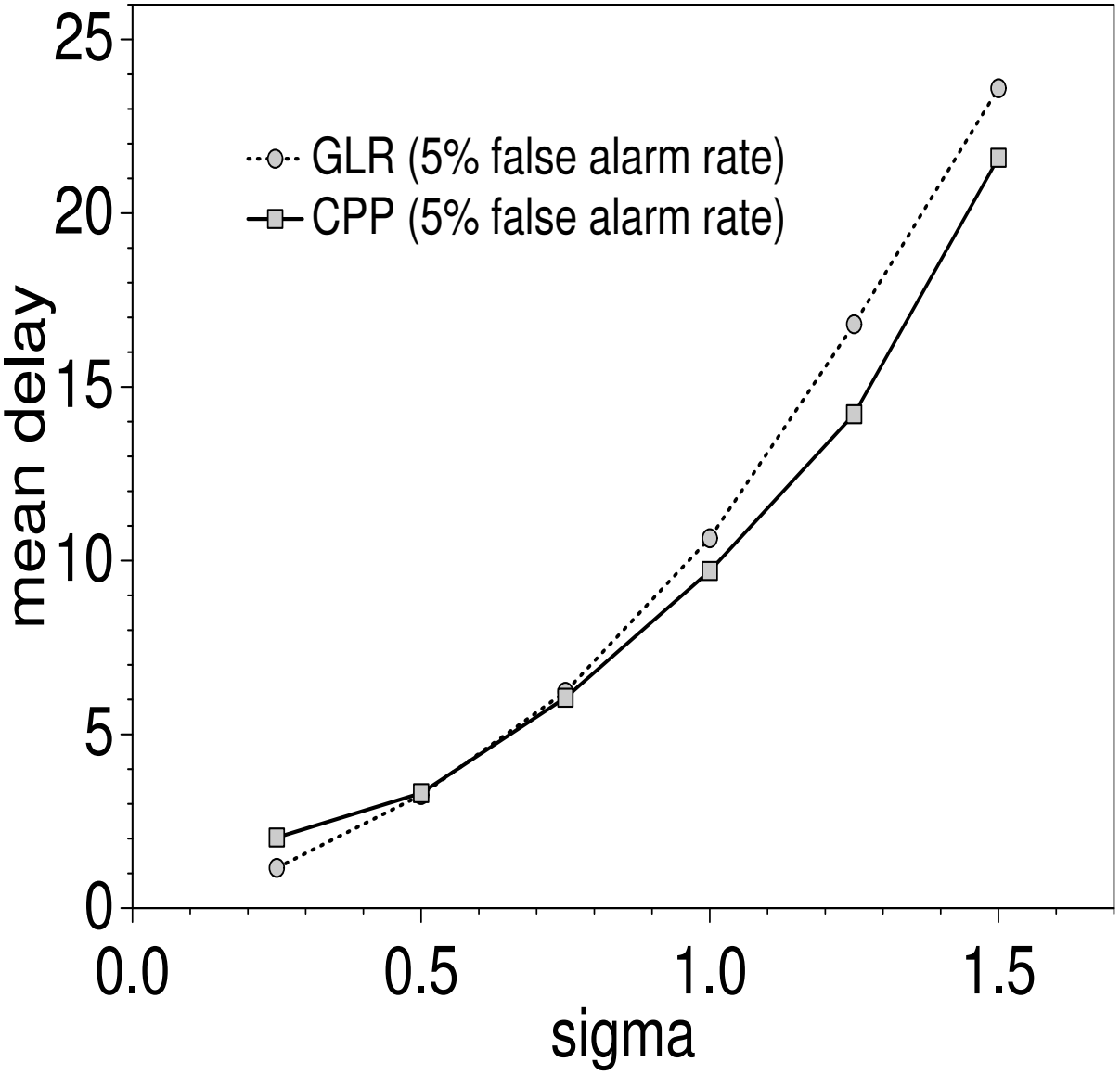
CPP = change point probability

$$P(changepoint) = \sum_{i=0}^{k} P(i^{+})$$

- Alarm when $P(changepoint) > thresh$.



- $\mu = 0, 1$
- $\sigma = 1$
- $\tau \sim \text{Exp}(0.01)$
- Goodness = lower mean delay for same false alarm rate.



- Fix false alarm rate = 5%
- Vary σ.
- CPP does well with small S/N.

So it works on a simple problem.

Future work:

- Other changepoint problems (location, tracking, prediction).
- Other data distributions (lognormal).
- Testing robustness (real data).

Good news:

- Very general framework.
- Seems to work.
- Many possible applications.

Bad news:

- Still some wrinkles to iron.
- n^2 space and time may be fatal.
- May be overkill for original application.

- More at
  `http://allendowney.com/changepoint`
- Or email `downey@allendowney.com`